Linux

- What is server and why we need infrastructure
  - Server is nothing but it serves the information
  - If we want to host some application we may require infra for ex: serves, cables,ram,hdd,network …etc
  - AWS company providing infrastructure over the web , we can get infra from aws
  - Aws serving the servers as pay as you go model
  - In 2006 aws began offering It services to business , aws providing more than 200+ countries

Aws Providing services across the globe using regions and availability zones .

  - Region means geographical region
  - Availability xone means a room with “data center “ having with complex networks.

it is providing services for 190+ countries , we have 26 regions and 84 A.Z

If some power shortage or disaster occurred , then other a.z will be available to host

- Aws always thinks to have backup
- Its highly recommended to keep our servers in multiple a.z
- If one a.z goes down oue servers will be running in other a.z

Windows:

- Windows o.s provided by Microsoft company
- Windows o.s is a paid software (we have to purchase)
- Windows is a single user based o.s (only one user can perform)
- One user can run multiple apps at a time
- Install antivirus to protect our app from virus manually
- It is having beautiful gui

Linux :

- To overcome the problems we will use linux
- Its an opensource operating system
- It’s a multiuser based operating system , very secured and community based o.s
- Linux o.s is highly recommended for business applications to run servers app hosting , db severs.

Evolution of Linux:

- First o.s came into market in 1950’s
- Bell labs , General motor research lab implemented the first o.s in the early 1950’s
- In 1960 the version of unix came into market

- In 1991 Linux torvarlds made some changes and released into the market for free of cost , that is called linux o.s

Linux Distributions :

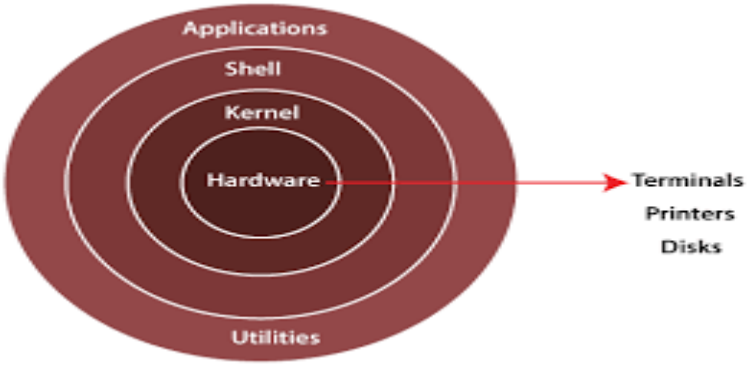
- Open source anybody can take and customized according to their requirement and released into the markey with different names.
- So many companies and people had taken the opensource code and customized according to their requirements and released into the market with dirrenet names
- For ex: redhat , Ubuntu , centos , fedora, susu-linux.

What is Operating system:

- An Operating system is a software that acts as a mediator between computer hardware and the end user.

What is Linux O.S Architecture

It has several layers : hardware , kernel , shell and application utilities



- Hardware Consist of all the physical devices attached to the system for ex: hd , ram , cpu ..etc
- Kernal is the heart of the o.s , which will interact with hardware directly
- Shell is the interface between the user and kernel , takes input from user and gives to the kernak , and takes output from kernel and it gives te output to the user
- Application utilities are the utilities which runs on the shell for ex: browser , text editor ..etc.

Linux File system:

- **Everything in the linux will be treated as file only**

- All the files in the linux o.s are divided into three types

    1. Ordinary files 2. Directory files 3. Device files

1. Ordinary files nothing but the files which contains data directory are called ordinary files For ex: Text files , binary files ..etc.
2. Directory files : Directory files in linux nothing but Folders in windows ex: devops, any directory
3. Device files will be represented as a file by using device files we can communicate with the devices.

How to check file type in Linux

- ls –l , to display all files with its file type we use ls –l
- After giving the first char of the output represent file type
- D – firectory
- - file
- L – link file

Linux file system hierarchy:

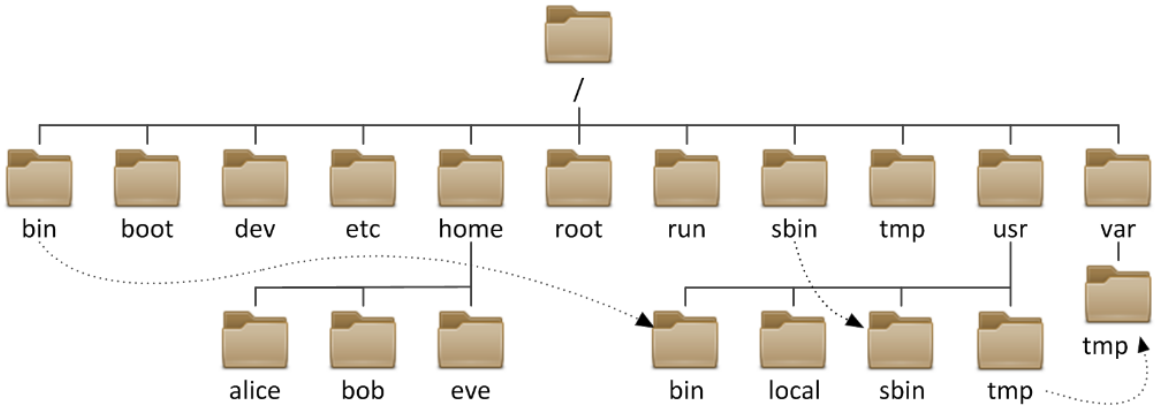
Root is the entry point of the file system

/ - indicated the root directory

Root – is the home directory of root

= There are some subfolders in the root directory

1. /home: contains user info details , user home directories
2. /bin : contains binaries for commands
3. /boot: some boot loader files present in this directory
4. /dev: device files will be there in this directory
5. /etc: Host –specific configuration files will be present in this directory
6. /lib: libraries for binaries containes here
7. /media: Mount point for removable devices
8. /mnt : Temporary mounted file system
9. /opt: Third party softwares we install here
10. /sbin : system binaries contains executable binary files
11. Tmp : temporary file , files will be deleted once we reboot the system
12. /usr : user utilities and applications



Linux Commands

ls

- **Meaning**: List directory contents.
- **Explanation**: It shows you a list of files and directories in the current directory where you run the command

- **Example**: `ls -l` (Lists files in long format)
- **ls** –t : it displays files based on last modified date & time.
- **Ls** –a : shows hidden files
- **Ls** – li : it displays the inode number , address of the file.

cd

- **Meaning**: Change directory.
- This command is used to change the current working directory in the terminal

- **Example**: `cd /home/user` (Changes to /home/user directory)
- **Cd ..** = To come out from one directory
- **Cd -** = To point the previous state

pwd

- **Meaning**: Print working directory.
- **Explanation**: This command tells you the full path to the directory you are currently

- **Example**: `pwd` (Displays the current directory path)

cp

- **Meaning**: Copy files and directories.
- This command is used to copy files or directories from one location to another

- **Example**: `cp file1.txt /home/user` (Copies file1.txt to /home/user)

Date : which is going to print Current date

Explanation: This command displays the current date and time according to your system's clock.

Example: date

- Date +%Y , date+%m , date +%d

Cal : Print the current month calender

Explanation: This command displays a calendar for the current month

Example: cal

- Cal 06 2000- it prints june 2000 calender

Mv : Move/rename files and directories & move the files ad dir to another dir .

Explanation: This command is used to move files or directories from one location to another. It also serves the purpose of renaming files or directories

- **Example**: `mv file1.txt file2.txt` (Renames file1.txt to file2.txt)
- mv file1.txt /path/to/new/location/

WC : we can count file content data with “wc” command

Explanation: This command is used to count the number of lines, words, and characters in a file

- Example: wc states.txt
- Output: lines words characters <filename>

rm

- **Meaning**: Remove files or directories.
- **Explanation**: This command is used to remove or delete files or directories

- **Example**: `rm file1.txt` (Deletes file1.txt)
- rm –rf < directory name>

touch

- **Meaning**: Create an empty file or update file timestamps.
- **Explanation**: The `touch` command is used to create empty files

Example: `touch newfile.txt` (Creates a new empty file named newfile.txt)

mkdir

- **Meaning**: Make directories.
- **Explanation**: The `mkdir` command is used to create new directories (folders) in the filesystem.

- **Example**: `mkdir newdir` (Creates a directory named newdir)
- mkdir -p /path/to/new/directory - creates a nested directory
- mkdir -m 755 newdirectory - create a directory with specific permissions

rmdir

- **Meaning**: Remove empty directories.
- **Explanation**: The `rmdir` command is used to remove or delete empty directories in the filesystem
- **Example**: `rmdir olddir` (Deletes an empty directory named old dir)

ln

- **Meaning**: Create links between files.
- **Explanation**: The `ln` command is used to create links between files. There are two types of links it can create:

- **Example**: `ln -s /path/to/file linkname` (Creates a symbolic link to a file)
- ln /path/to/targetfile linkname - Hard link - They share the same inode and have the same permissions and file contents.

cat

- **Meaning**: Concatenate and display file content.
- **Explanation**: The `cat` command is used to display the contents of one or more files to the standard output (usually the terminal). It can also be used to concatenate (combine) multiple files into a single file or display file contents sequentially

- **Example**: `cat file.txt` (Displays the content of file.txt)
- **Cat > file1** : creates file with content
- **Cat >> file1** : to append the content

less

- **Meaning**: View file contents one screen at a time.
- The `less` command is used to view the contents of a file interactively, allowing you to scroll up and down through the file

- **Example**: `less file.txt` (Opens file.txt in the less viewer)

more

- **Meaning**: View file contents one screen at a time (less feature-rich than `less`).
- The `more` command in Linux is similar to `less` in that it allows you to view the contents of a file one screenful at a time. It's used for paging through text, but unlike `less`, it does not support scrolling backwards in the file

- **Example**: `more file.txt` (Displays file.txt contents page by page)

head

- **Meaning**: Output the first part of files.
- **Explanation**: The `head` command is used to display the first few lines (by default, the first 10 lines)
- **Example**: `head -n 10 file.txt` (Shows the first 10 lines of file.txt)

tail

- **Meaning**: Output the last part of files.
- **Explanation**: The `tail` command is used to display the last few lines (by default, the last 10 lines)
- **Example**: `tail -n 10 file.txt` (Shows the last 10 lines of file.txt)

echo

- **Meaning**: Display a line of text.
- **Explanation**: The `echo` command is used to display a line of text or the value of a variable.
- **Example**: `echo "Hello, World!"` (Prints "Hello, World!" to the terminal)

Grep : Grep is a powerful command which is used to serach a string or word in a file with matches the pattern

- **Meaning**: Print lines matching a pattern.
- **Explanation**: The `grep` command is used to search for lines matching a specified pattern in one or more files or input

- **Example**: `grep "pattern" file.txt` (Searches for "pattern" in file.txt)
- Grep –i linux unix.txt -i case-sensitive

- grep –c Andhra states.txt -c : count the lines

diff **:** To compare files on linux

Explanation: The `diff` command is used to compare the contents of two files and display the differences between them.

example: diff file1.txt file2.txt

sed

- **Meaning**: Using sed command we can perform operations without opening the file , suppose I want to replace a word in the file wherever the word present in the file.

Explanation: The `sed` command is used to perform text transformations on input streams or files using regular expressions.

- **Example**: `sed 's/old/new/g' file.txt` (Replaces all occurrences of "old" with "new" in file.txt)
- Sed ‘s/unix/LINUX/g’ Linux.txt
- Sed ‘3d’ file.txt – delete particular line in the file

Man : man command is used to get documentation of particular command

Explanation: The `man` command is used to display the manual pages for other commands,

Example: man ls

find

- **Meaning**: Search for files in a directory hierarchy.

Explanation: The `find` command is used to search for files and directories based on various criteria such as name, type, size, permissions, and more.

- **Example**: `find /home -name "*.txt"` (Finds all .txt files in /home)
- **Find /tmp –type** –f –empty
- **Find / tmp –type** –d –empty
- find . -type f -mtime +7 -exec rm {} \;

locate

- **Meaning**: Find files by name.
- **Explanation**: The `locate` command is used to quickly locate files by name in a pre-built database (updated regularly by the `updatedb` command).

- **Example**: `locate file.txt` (Finds the file.txt in the file system)

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vmstat

- **Meaning**: Report virtual memory statistics.
- **Explanation**:The `vmstat` command reports information about processes, memory, paging, block IO, traps, and CPU activity
- **Example**: `vmstat 1 5` (Displays virtual memory stats every second for 5 intervals)

du

- **Meaning**: Estimate file space usage.
- **Explanation**: The `du` command is used to estimate file space usage. It summarizes disk usage of directories and files recursively
- **Example**: `du -sh /home/user` (Displays the disk usage of /home/user in human-readable format)

df

- **Meaning**: Report file system disk space usage.
- **Explanation**: The `df` command is used to display information about the disk space usage on mounted file systems
- **Example**: `df -h` (Shows disk space usage in human-readable format)

ps

- **Meaning**: Report a snapshot of current processes.
- **Explanation**: The `ps` command is used to display information about currently running processes on a system. It provides a snapshot of processes

- **Example**: `ps aux` (Displays all running processes)
- ps auxf - Display all processes in a tree-like format:

top

- **Meaning**: Display Linux tasks.
- **Explanation**: The `top` command provides a dynamic real-time view of system processes. It displays information about CPU usage, memory usage, running processes, and more.
- **Example**: `top` (Shows the system's activity information in real-time)

htop

- **Meaning**: Interactive process viewer (similar to top).
- **Explanation**: `htop` is an interactive process viewer for Unix systems, providing a real-time overview of system resources and processes.

- **Example**: `htop` (Opens the htop interactive process viewer)

tee

- **Meaning**: Read from standard input and write to standard output and files.
- **Explanation**: The `tee` command reads from standard input and writes to standard output and one or more files simultaneously. It is often used in pipelines to split the output of a command

- **Example**: `echo "Hello" | tee file.txt` (Writes "Hello" to file.txt and displays it)

sort

- **Meaning**: Sort lines of text files.
- **Explanation**: The `sort` command is used to sort lines of text files or streams of text data. It arranges lines alphabetically or numerically, depending on the options used.
- **Example**: `sort file.txt` (Sorts the lines in file.txt alphabetically)

uniq

- **Meaning**: Report or omit repeated lines.
- **Explanation**: The `uniq` command is used to filter out repeated lines in a sorted file or stream of data.
- **Example**: `uniq file.txt` (Removes duplicate lines from file.txt)

tr

- **Meaning**: Translate or delete characters.

- **Example**: `echo "hello" | tr 'a-z' 'A-Z'` (Converts lowercase letters to uppercase)

cut

- **Meaning**: Remove sections from each line of files.
- **Explaination**: The `cut` command is useful for extracting specific columns or characters from structured text data, making it easier to manipulate and process text files

- **Example**: `cut -d':' -f1 /etc/passwd` (Displays the first field of each line in /etc/passwd/
- Cut –b 1-4 filename

diff

- **Meaning**: Compare files line by line.
- **Example**: `diff file1.txt file2.txt` (Shows differences between file1.txt and file2.txt)**wc**
- **Meaning**: Print newline, word, and byte counts for each file.
- **Example**: `wc -l file.txt` (Counts the number of lines in file.txt)

kill

- **Meaning**: Send a signal to a process.
- **Explanation**: The `kill` command is used to terminate processes by sending a signal to a specified process ID (PID) or job.

- **Example**: `kill 1234` (Sends the default SIGTERM signal to process ID 1234)

killall

- **Meaning**: Kill processes by name.
- **Explanation**: The `killall` command is used to terminate processes by their name instead of their PID. It sends signals to all processes with the specified name.

- **Example**: `killall firefox` (Kills all processes named firefox)

pkill

- **Meaning**: Kill processes by name.

Explanation: Similar to `killall`, `pkill` kills processes based on matching criteria, which can include partial names, command lines, or other patterns.

-
- **Example**: `pkill -9 firefox` (Forcefully kills all processes named firefox)

service

- **Meaning**: Run a System V init script.
- The `service` command is used to start, stop, restart, and manage services on Unix-like systems using init scripts. It's commonly used on older Linux distributions.
- **Example**: `service apache2 start` (Starts the Apache2 service)

systemctl

- **Meaning**: Control the systemd system and service manager.
- **Explanation**: `systemctl` is a more modern command used to manage systemd services and units on Linux distributions that use systemd
- **Example**: `systemctl restart apache2` (Restarts the Apache2 service)

journalctl

- **Meaning**: Query and display messages from the journal.
- **Explanation**: `journalctl` is a command-line utility for querying and displaying logs from the systemd journal
- **Example**: `journalctl -u apache2` (Displays logs for the Apache2 service)

nohup

- **Meaning**: Run a command immune to hangups, with output to a non-tty.
- `nohup` stands for "no hang up". It's a command used in Unix-like operating systems (like Linux) to run other commands or scripts in a way that they continue running even after you log out or close your terminal session.

- **Example**: `nohup command &` (Runs command in the background, immune to hangups)

free

- **Meaning**: Display memory usage.
- **Explanation**: The `free` command is used to display information about the total amount of free and used physical and swap memory in the system
- **Example**: `free -h` (Shows memory usage in human-readable format)

ssh

- **Meaning**: OpenSSH SSH client (remote login program).
- **Explanation**: The `ssh` command is used to establish a secure, encrypted connection to a remote server or computer over an unsecured network. It provides a secure way to access and manage remote systems.

- **Example**: `ssh user@remote` (Logs into the remote server)
- ssh –I pem-key path username@ip/dns
- ssh -i ~/.ssh/id_rsa username@remote_host

System level commands

Uname : The `uname` command in Linux is used to display system information

Example: uname

- Uname –a : Display all system information
- Uname –r : Display kernel release
- Uname –v : Display kernel version

which

- **Meaning**: Show the full path of (shell) package.
- **Example**: `which <package name>`

Update:

- Refreshes the local list of available packages from the repositories, ensuring you have the latest information about package versions.

Upgrade:

- Installs newer versions of the packages currently installed on your system, ensuring your software is up-to-date with the latest versions available in the repositories.

apt:

- **Meaning**: Package manager for Debian-based Linux distributions like Ubuntu.

dnf:

- **Meaning**: Package manager for Red Hat-based Linux distributions like Fedora.

rpm:

- **Meaning**: Package manager and package format for Red Hat-based Linux distributions.

pacman:

- **Meaning**: Package manager for Arch Linux and its derivatives.

uptime

- **Meaning**: Tell how long the system has been running.
- **Example**: `uptime` (Shows how long the system has been up)

Who : The `who` command in Linux is used to display information about users who are currently logged into the system

Example: who

- Who –a : Display all information, including IP addresses and idle times

whoami

- **Meaning**: Print the current user's username.
- **Example**: `whoami`

id : The `id` command in Linux is used to display user and group information for a specified user

Example: id <username>

USER MANAGEMENT COMMANDS

Useradd : we use this command to create the user

Example: sudo useradd –m vaseem

- Sudo adduser vaseem

Su : Switching the user in linux

Example: su username

Userdel : to delete the user

Example: sudo userdel vaseem

- Sudo deluser –remove-home user1

Groupadd **:** to create the group

Eample: sudo groupadd devops

Usermod : Adding user to the group

Example: sudo usermod -a –G <groupname> <username>

Gpasswd : to delete the user from group

Eample: sudo gpasswd –d <usernma> <groupname>

Groupdel : to delete the group

Example: sudo groupdel <groupname>

File Permission Commands

chmod

- **Meaning**: We use files to store the data inorder to protect our file we use file permissions

  We should not allow everyone to access config files so I want to give permission only for specific user for that I use this concept

- **Example**: `chmod 755 script.sh` (Sets permissions to rwxr-xr-x for script.sh)

chown

- **Meaning**: Change file owner and group.
- **Example**: `chown user:group file.txt` (Changes the owner to user and the group to group for file.txt)
- Chgrp vaseem file.txt – changing the group of file

Compression Commands

File Transfer commands

rsync

- **Meaning**: Remote file and directory synchronization.
- **Example**: `rsync -avz /path/to/source user@remote:/path/to/destination` (Synchronizes directories between local and remote)

scp

- **Meaning**: Secure copy (remote file copy program).
- **Example**: `scp file.txt user@remote:/path/to/destination` (Copies file.txt to a remote server)
- Scp –i key-path <file/dir pat> username@ip/dns:path-to-destination

Linux Networking –Commands

ping

- **Meaning**: Send ICMP ECHO_REQUEST to network hosts.
- **Example**: `ping google.com` (Pings the google.com server)

netstat

- **Meaning**: Print network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.
- **Example**: `netstat -tuln` (Displays active listening ports)

ifconfig

- **Meaning**: Configure a network interface.
- **Example**: `ifconfig eth0 up` (Brings up the eth0 network interface)

traceroute

- **Meaning**: Print the route packets take to the network host.
- **Example**: `traceroute google.com` (Traces the route to google.com)

tracepath

- **Meaning**: Traces the path packets take to a network host.
- **Example**: `tracepath [destination]`

mtr

- **Meaning**: Network diagnostic tool combining `ping` and `traceroute`.
- **Example**: `mtr [destination]`

nslookup

- **Meaning**: Query Internet name servers interactively.
- **Example**: `nslookup google.com` (Looks up the IP address of google.com)

telnet

- **Meaning**: Communicate with another host using the TELNET protocol.
- **Example**: `telnet [hostname] [port]`

hostname

- **Meaning**: Show or set the system's host name.
- **Example**: `hostname` (Displays the current hostname)

ip

- **Meaning**: Show/manipulate routing, devices, policy routing, and tunnels.
- **Example**: `ip addr show` (Shows all IP addresses on the system)

iwconfig

- **Meaning**: Configure wireless network interfaces.
- **Example**: `iwconfig [interface] [options]`

Usage Examples:

1. **Display wireless network information for all interfaces:**

   ```
   Iwconfig
   ```

ss

- **Meaning**: Utility to investigate sockets.
- **Example**: `ss -tuln` (Displays listening TCP and UDP sockets)

dig

- **Meaning**: DNS lookup.
- **Example**: `dig google.com` (Performs a DNS lookup for google.com)

whois

- **Meaning**: Query the WHOIS database for information about domains and IP addresses.
- **Example**: `whois [domain]`

nc

- **Meaning**: Utility for managing network connections.
- **Example**: `nc [options] [hostname] [port]`

arp

- **Meaning**: Manipulate the system ARP cache.
- **Example**: `arp -a` (Displays the ARP cache)

ifplugstatus

- **Meaning**: Display link status of network interfaces.
- **Example**: `ifplugstatus [options] [interface]`

curl

- **Meaning**: Transfer data from or to a server.
- **Example**: `curl -O http://example.com/file.zip` (Downloads file.zip from example.com)

wget

- **Meaning**: Non-interactive network downloader.
- **Example**: `wget http://example.com/file.zip` (Downloads file.zip from example.com)

route

- **Meaning**: Show/manipulate the IP routing table.
- **Example**: `route -n` (Displays the IP routing table)

iptables

- **Meaning**: Administration tool for IPv4 packet filtering and NAT.
- **Example**: `iptables -L` (Lists all current iptables rules)

watch

- **Meaning**: Execute a program periodically, showing output fullscreen.
- **Example**: `watch [options] [command]`

nmap

- **Meaning**: Network exploration tool and security/port scanner.
- **Example**: `nmap [options] [target]`

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iostat

- **Meaning**: Report CPU and I/O statistics.
- **Example**: `iostat -x` (Shows extended I/O stats)

dmesg

- **Meaning**: Print or control the kernel ring buffer.
- **Example**: `dmesg | grep error` (Displays kernel messages containing "error")

ftp

- **Meaning**: Internet file transfer program.
- **Example**: `ftp ftp.example.com` (Connects to an FTP server)

sftp

- **Meaning**: Secure File Transfer Protocol.
- **Example**: `sftp user@remote` (Connects to a remote server via SFTP)

lsblk

- **Meaning**: List information about block devices.
- **Example**: `lsblk` (Lists all block devices)

firewalld

- **Meaning**: A firewall management tool.
- **Example**: `firewall-cmd --state` (Checks the state of firewalld)

chkconfig

- **Meaning**: System V runlevel configuration tool.
- **Example**: `chkconfig httpd on` (Enables the httpd service at boot)

crontab

- **Meaning**: Schedule periodic jobs.
- **Example**: `crontab -e` (Edits the current user's cron jobs)

at

- **Meaning**: Schedule commands to be executed at a later time.
- **Example**: `at 10:00` (Schedules a job to run at 10:00)

jobs

- **Meaning**: List active jobs.
- **Example**: `jobs` (Displays all active jobs)

bg

- **Meaning**: Place a job in the background.
- **Example**: `bg %1` (Places job 1 in the background)

fg

- **Meaning**: Place a job in the foreground.
- **Example**: `fg %1` (Brings job 1 to the foreground)

alias

- **Meaning**: Create an alias for a command.
- **Example**: `alias ll='ls -l'` (Creates an alias ll for `ls -l`)

unalias

- **Meaning**: Remove an alias.
- **Example**: `unalias ll` (Removes the alias ll)

history

- **Meaning**: Show command history.
- **Example**: `history` (Displays command history)

man

- **Meaning**: An interface to the system reference manuals.
- **Example**: `man ls` (Displays the manual page for the ls command)

whatis

- **Meaning**: Display one-line manual page descriptions.
- **Example**: `whatis ls` (Displays a one-line description of the ls command)

apropos - **Meaning**: Search the manual page names and descriptions.

Example: `apropos network` (Searches for commands related to "network")

awk

- **Meaning**: Pattern scanning and processing language.

- **Example**: `awk '{print $1}' file.txt` (Prints the first column of file.txt)

xargs

- **Meaning**: Build and execute command lines from standard input.
- **Example**: `find . -name "*.txt" | xargs rm` (Finds all .txt files and deletes them)